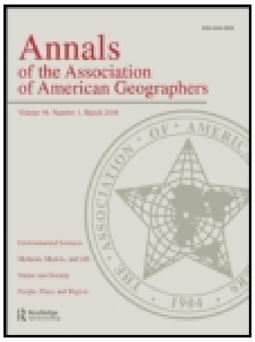

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Uncertainty Analysis for a Social Vulnerability Index

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Indexes have gained favor over the past decade as a tool to measure social vulnerability to hazards. Numerous index designs have been put forward, yet we still know very little about their reliability. This research investigates the methods of social vulnerability index construction, examining decisions related to indicator selection, scale of analysis, measurement error, data transformation, normalization, and weighting. Each of these stages is imbued with uncertainty due to choices made by the index developer. The study applies Monte Carlo–based uncertainty analysis to assess and visualize uncertainty for a hierarchical social vulnerability index. Confidence limits are computed for the index rankings, leading to a finding of a high magnitude of uncertainty. The performance of the index compared to alternative configurations is strong in some places but statistically biased in about a third of the census tracts. The variability of index rankings is also assessed, indicating that index precision decreases with increasing vulnerability. Uncertainty analysis provides a useful, yet largely unapplied stage of index production that highlights places where the model is most reliable. If applied to the creation of social vulnerability indexes, output metrics can be produced with a greater degree of precision, transparency, and credibility. *Key Words: hazards, index, indicators, social vulnerability, uncertainty.*

过去十多年来，索引做为衡量社会之于灾害的脆弱性之工具受到青睐。大量的索引设计出炉，但我们对于它们的可信度却所知无多。本研究探讨社会脆弱性索引的建构方法，检视与索引选择、分析尺度、测量错误、数据传输、规范化与加权的相关决定。上述每个步骤都因索引制作者的选择而带有不确定性。本研究运用基于蒙地卡罗的不确定性分析，评估并视觉化阶层体系的社会脆弱性索引之不确定性。在推估索引排序的信赖界线后，发现高度的不确定性。相较于其他配置而言，该索引的效能在某些部分较为坚实，但却约有三分之一的普查范围具有统计偏见。本文同时关注索引排名的变异性，并指出索引的准确性随着脆弱性的增加而降低。不确定性分析提供了索引生产中能突显模式最可信赖处、却多半尚未运用的有用阶段。若将之用运于社会脆弱性指标的设计，将可生产具有较大的准确性、透明度以及可信度的产出衡量标准。*关键词：灾害，索引，指针，社会脆弱性，不确定性。*

Durante la pasada década, los números índices han ganado preferencia como herramientas para medir la vulnerabilidad social al riesgo. Se han propuesto numerosos diseños de índices, aunque todavía conocemos poco acerca de su confiabilidad. Este estudio investiga los métodos de construcción de índices de vulnerabilidad social, examinando las opciones que tienen que ver con la selección de indicadores, escala de análisis, medidas del error, transformación de datos, normalización y medida del peso. Cada una de estas etapas lleva implícita la incertidumbre generada por las decisiones que toma quien desarrolla el índice. El estudio aplica un análisis de incertidumbre basado en Monte Carlo para evaluarla y visualizarla en un índice de vulnerabilidad social jerarquizada. Se calcularon los límites de confianza para la clasificación del índice, lo que llevó al hallazgo de una alta magnitud de incertidumbre. El desempeño del índice en comparación con configuraciones alternativas es fuerte en algunos lugares pero aparece estadísticamente sesgado en alrededor de un tercio de los distritos censales. También se evaluó la variabilidad de las clasificaciones del índice, lo cual indica que la precisión del índice disminuye a medida que se incrementa la vulnerabilidad. El análisis de incertidumbre pone de manifiesto una útil aunque hasta ahora poco aplicada etapa de producción de índices que relieva los lugares donde el modelo es más confiable. Si son aplicadas a la creación de índices de vulnerabilidad social, las métricas de salida pueden producirse con un gado mucho mayor de precisión, transparencia y credibilidad. *Palabras clave: riesgos, índices, indicadores, vulnerabilidad social, incertidumbre.*

Hazards are increasingly being examined through the lens of social vulnerability, which examines how political, social, economic, and institutional factors lead some population groups to disproportionately suffer adverse natural hazard impacts. Through social science research conducted largely over the past two decades, the social and institutional norms that produce social vulnerability are now better understood. Transforming broad understanding of vulnerability processes into formalized strategies for its reduction

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requires a clear comprehension of the identity and capacities of vulnerable groups, however. Vulnerability assessment is an important tool for gaining this understanding, as well as assigning mitigation priorities and monitoring progress (Downing et al. 2001). The idealized assessment process begins with a qualitative characterization of the study area, followed by identification of vulnerability drivers, development of a quantitative vulnerability model, and communication of findings with stakeholders (Polsky, Neff, and Yarnal 2007). The quantification stage is typically achieved by modeling social vulnerability through the construction of vulnerability indexes, which serve to simplify the multidimensional complexity of social vulnerability into a single metric.

Typical vulnerability index development involves sequential stages including the selection of demographic indicators, normalization of indicators to a common scale, and summation to a final value. As with any model, however, changes in input data and algorithms have the potential to significantly influence the output. Although there is broad interest in the need to quantitatively model social vulnerability, there is far less consensus regarding the ideal set of methods used for the production of indexes. This lack of consensus means that uncertainty is introduced into the modeling process whenever an index developer chooses between competing viable options. Due to this subjectivity, current vulnerability indexes could be flawed and thus fail to reflect the reality they seek to convey (Barnett, Lambert, and Fry 2008). U.S. agencies at state and federal levels (California Office of Emergency Services 2010; Dunning and Durden 2011) have begun to take a greater interest in the use of social vulnerability indexes in the hazard mitigation planning process. Decisions regarding project prioritization, target areas, and resource allocation might be flawed, however, if they rely on index rankings with an unknown degree of uncertainty. A first step toward understanding the impact of the subjectivity in social vulnerability modeling is to measure the uncertainty it produces in resultant indexes.

Uncertainty

The assessment of uncertainty in models of complex systems has long been an area of research in the physical sciences and engineering, with the underlying goal of placing error bounds on model outputs. Uncertainty information is now widely reported in model predictions of atmospheric warming, sea level rise, and hurricane paths (e.g., the cone of uncertainty). Models are now understood to be associated with two general forms of uncertainty. The first is aleatoric, which occurs due to heterogeneity or the inherent randomness of input parameters and processes. The term is of Latin origin, meaning "rolling of dice" (Kiureghian and Ditlevsen 2009). As an example of aleatoric uncertainty, consider the timing of a hazardous event such as an earthquake. There is variability in the event timing in that it can just as easily occur in the middle of the night, during rush hour, or on the weekend. The social impact of the quake depends greatly on this timing, and therefore the output of an earthquake impact model can also be subject to aleatoric uncertainty. Aleatoric uncertainty is considered to be essentially irreducible and in models is often represented using a probability distribution for the input parameter of interest. Epistemic is the second uncertainty form; it stems from an incomplete or imprecise understanding of parameters that are modeled with fixed but poorly known values (Helton et al. 2010). Epistemic uncertainty in models can arise from the use of expert opinion or misrepresentation of processes (Jakeman, Eldred, and Xiu 2010). It follows that with improved knowledge, data, and modeling processes, epistemic uncertainty can be reduced.

Aleatoric uncertainty might affect the input data used for social vulnerability indexes, but epistemic uncertainty accompanies every stage of index construction. These stages can include conceptual framework development, selection of indicators, collection of data, consideration of measurement error, imputation of missing data, data transformation, scaling, weighting, and aggregation (Organization for Economic Co-operation and Deveopment [OECD] 2008). As modeling decisions during the development of the index proceed, epistemic uncertainty associated with each step propagates and potentially interacts with that of previous steps. To fully understand the effect of multiple and potentially interacting modeling stages, uncertainty and sensitivity analysis are required. Uncertainty analysis helps assess the robustness of index ranks, and global sensitivity analysis decomposes the uncertainty to determine which index construction decisions have the greatest influence on the output rank variability. The analyses are coupled and apply Monte Carlo simulation to generate a frequency distribution of index ranks for each enumeration unit through the computation of reasonable alternative model configurations. This research focuses on uncertainty analysis, with the goal of characterizing the magnitude and spatial distribution of epistemic uncertainty in the modeling of a social vulnerability index. Uncertainty analysis has previously been used in the evaluation of indexes of disaster risk (Marulanda, Cardona, and Barbat 2009) and sustainability (Munda et al. 2009). The approach has not been systematically applied to the development of indexes of social vulnerability to hazards, however.

Three questions form the basis for this work:

1. How much uncertainty is associated with the social vulnerability index?
2. How does the index perform when alternative configurations are considered?
3. What is the relationship between vulnerability and uncertainty?

To address these questions, design options for a hierarchical social vulnerability index for Sarasota County, Florida, are subjected to an uncertainty analysis using Monte Carlo simulation. The article proceeds as follows. The first section describes the major sources of epistemic uncertainty in the index development process. The second section details the methods through which the uncertainty sources were modeled. The third section explains the findings, exploring the index ranks in terms of uncertainty magnitude, bias, and precision. The final section concludes with ways in which uncertainty analysis can be used to improve social vulnerability indexes.

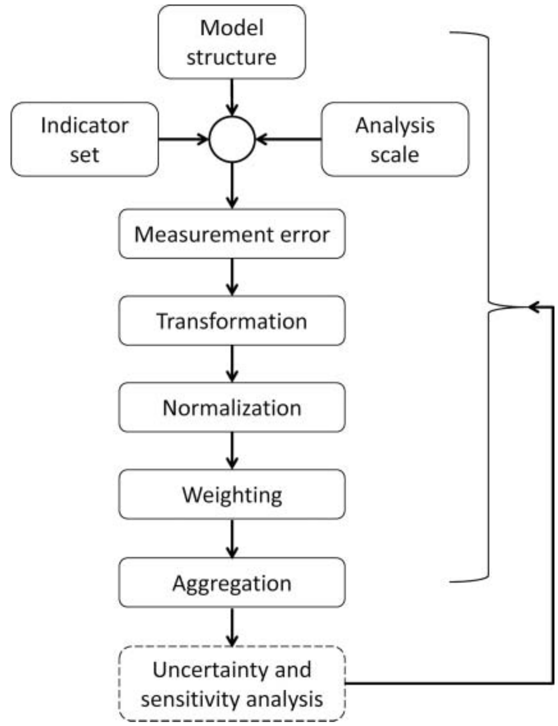

Figure 1. Index construction flowchart.

Sources of Uncertainty

Uncertainty in social vulnerability indexes stems from modeling decisions made during their development. Vulnerability index construction is largely a sequential process; the primary stages of development are shown in Figure 1. Ideally, uncertainty and sensitivity analysis should make up the final stage, with its findings used to identify and ultimately reduce the key sources of uncertainty in index assumptions. With indexes of social vulnerability to hazards, however, this final step is rarely implemented. The modeling begins with a choice of model structure. The structures employed by social vulnerability index developers generally fall into one of three categories. Deductive designs (Cutter, Mitchell, and Scott 2000; Lein and Abel 2010) consist of up to ten normalized variables that are assembled to compute the index. Hierarchical models (Flanagan et al. 2011; Mustafa et al. 2011) use a greater number of indicators that are grouped into thematic subindexes, which are then combined to form the index. Inductive approaches (Fekete 2009; Finch, Emrich, and Cutter 2010) typically begin with more than twenty variables, which are reduced to a smaller number of latent variables using principal components analysis and aggregated to compute the index.

The goal of indicator selection is to choose proxy variables for the underlying theoretical dimensions of social vulnerability. These dimensions could include socioeconomic status, age, race and ethnicity, gender, disability, housing tenure, language proficiency, and health (Phillips et al. 2010). The particular dimensions represented in the index depend on the goal of the index. Is the index intended to apply to all hazards or is it hazard specific? Should it represent generalized vulnerability or focus on a particular stage of disaster progression (e.g., mitigation, warning, evacuation, response, recovery)? The statistical properties of the indicators can also play a role in their inclusion, as highly correlated variables within the same dimension are not desirable. Finally, practical considerations such as data availability and cost might drive the indicator selection process.

Concurrent with indicator selection is a choice of analysis scale. Consideration of the scale of analysis of social data has long received a great deal of attention in geographical analysis. The modifiable areal unit problem can affect geographical studies that employ aggregated data. It manifests as a combination of a scale problem, where correlations between variables often increase with the level of aggregation, and an aggregation problem, in which correlations between enumeration units might depend as much on the manner of the aggregation as the relationships between variables (Openshaw and Taylor 1983). As a result, in some cases it is not possible to aggregate or downscale indicators and still make valid conclusions (Villagrán de León 2006).

Error in the measurement of input data is another potential source of model uncertainty and one that has not been considered in previous vulnerability indexes. Potentially affecting social vulnerability indicators, the U.S. Census Bureau has long acknowledged chronic data collection error as a result of undercounts and overcounts (J. R. Clark and Moul 2003). Undercounts have historically been most severe with respect to racial and ethnic minorities, children, renters, migrant farm workers, the homeless, and undocumented immigrants (Hannah 2001; Passel 2005), which are among the demographic categories frequently included in vulnerability indexes.

The next step facing the index developer is how to transform the data. Although the dimensions of vulnerability have found widespread agreement among researchers, their representation has not. Is high vulnerability best characterized by the absolute number or relative proportion of vulnerable populations (Rygel, O'Sullivan, and Yarnal 2006)? In absolute representation, the counts of each variable are used, with high values indicative of high vulnerability. For example, if examining the indicator of elderly, one might use the total elderly count. An alternative representational method is relative composition, in which the indicator is computed as a proportion of its value in the analysis unit to its maximum potential value. Here the elderly indicator would be evaluated as percentage elderly, by dividing the elderly count by the total population. Areal density is another variable transformation option, in this case perhaps represented by the number of elderly per square kilometer. The choice between representational methods is critical because both absolute and relative size are important when evaluating vulnerability (Rygel, O'Sullivan, and Yarnal 2006). It has been found that using absolute size, the areas of greatest vulnerability will always be those with the greatest population (Jones and Andrey 2007). This technique is reasonable when the analysis units contain similar numbers of people so the index is not skewed by population size. Another potential application is when the index seeks to identify the number of vulnerable people, such as evacuation planning. Outside of these circumstances, ranks of vulnerability using absolute size might be functionally equivalent to the ranks of total population. With relative composition, the emphasis is on the proportion of vulnerable people. An enumeration unit with five vulnerable people out of ten would have the same transformed indicator value as one with 500 vulnerable out of 1,000. The use of relative composition allows for the comparison of vulnerability between rural and urban areas.

Variable normalization is the next development stage. Its purpose is to bring proxy variables with different measurement units and variability to a common, usually dimensionless scale (Rashed and Weeks 2003; Jones and Andrey 2007). In index construction, three broad normalization schemes have been typically applied: unscaled variables, linear scaling, and z score standardization. The use of unscaled variables is only appropriate when the data already exist in scaled form, such as indicators measured as percentages. Otherwise, it is important to scale the census variables because unscaled variables cannot be meaningfully combined (Booysen 2002; OECD 2008). Linear scaling is perhaps the most widely applied normalization technique. In min–max scaling, the difference between each value and the minimum value is divided by the range, generating values that fall between zero and one. Although min–max linear scaling is easy to comprehend, it does not perform well in the presence of outliers, as the resulting gulf between minimum and maximum values might distort index scores (Booysen 2002). In this case, linear scaling using only the maximum value might be preferred, calculating the transformed value by dividing the unscaled value by its maximum. Standardizing variables using z scores produces variables with a mean of zero and a standard deviation of one. Z score standardization is the preferred normalization approach for data sets with extreme values (Nardo et al. 2005). An added benefit of z score standardization is that the standardized variable has the same value regardless of the measurement unit of the raw variable (Nardo et al. 2005). Prior to any normalization, indicators for which vulnerability decreases as their values increase should undergo a directionality adjustment (Booysen 2002; Myers, Slack, and Singelmann 2008; Aguña and Kovacevic 2011). The adjustment is performed by taking either the additive (multiply by –1) or multiplicative (multiply by $1/x$) inverse. Inverse adjustment is especially applicable to vulnerability indicators of income, wealth, and medical access, for which high values are associated with low vulnerability.

The weighting of index components has been described as being among the most highly subjective decisions in the index construction process (Morse 2004; Böhringer and Jochem 2007). The goal is to bestow greater weights on those indicators deemed most important for measuring the phenomenon of interest (Booysen 2002). The derivation of unequal weights is often achieved through the use of either statistical or participatory approaches. Statistical weights might originate from metrics such as the percentage of variance explained in exploratory factor analysis, regression coefficients in linear regression, the inverse of the coefficient of variation, or data envelopment analysis. Most statistical methods are fairly easy to implement, but their data-driven focus might produce weights that do not conform to conceptual relationships between indicators. In practice, neither statistical nor participatory approaches have achieved primacy when assigning weights. That distinction goes to equal weights. Equal weighting could imply equivalent importance of each indicator or recognition that there is insufficient understanding of underlying processes to assign meaningful weights (Cutter, Boruff, and Shirley 2003; Nardo et al. 2005). Given the ubiquity of the use of equal weighting schemes, two points should be made. First, an index using equal weights is not an unweighted index. The use of differential weights is often criticized as too subjective, but the application of equal weights is no less so. Second, the existence of high correlations between indicators might introduce implicit weighting into an equal weighting scheme, as the associated dimensions could be effectively double counted.

The final stage of index development is aggregation. It refers to the method used to combine transformed, normalized, and weighted indicators into the final index. Commonly applied aggregation options include summation, multiplication, and multicriteria analysis. Additive aggregation is implemented in the index as a weighted linear combination of normalized indicators. The aggregate is often divided by the total number of indicators to compute the arithmetic mean. Largely owing to its simplicity and ease of understanding, summation is the most widely applied aggregation technique (Villa and McLeod 2002; Nardo et al. 2005). It also offers the benefit of insensitivity to outliers. An implicit assumption with an additive approach is that the normalized indicators have no influence on one another (Villa and McLeod 2002). This behavior, statistically known as *interaction*, means that the relationship between a group of indicators is constant across all values of each indicator. The assumption of no interaction can be particularly troublesome in social vulnerability analysis, in which the magnitude of several vulnerability dimensions has been observed to vary based on socioeconomic status (Phillips, Metz, and Nieves 2005; Elliott and Pais 2006). Another disadvantage of additive aggregation is that a low value in one indicator can mask a high value in another. This behavior is known as *compensability*. For example, an area with high levels of poverty might not receive a high vulnerability score if it also has low numbers of racial and ethnic minorities. To avoid issues with interaction and compensability, nonlinear aggregation techniques can be applied. Geometric aggregation is one such nonlinear approach and is implemented in the index as the product of normalized weighted indicators. Compensability is only partial in geometric aggregation. It tends to penalize enumeration units with greatly divergent normalized indicators and reward those with similar values (Aguña and Kovacevic 2011). In additive aggregation, compensability is constant, whereas compensability in geometric aggregation is low when there are indicators with low values (Nardo et al. 2005). A potential downside of geometric aggregations is a high sensitivity to outliers, but this feature might be desired if the identification of extreme index scores is intended. Additive and geometric approaches result in a quantitative index score, but multicriteria analysis methods employ nonlinear aggregation methods that generate index ranks instead of scores. These include approaches such as Pareto ranking and data envelopment analysis.

Study Area

Sarasota County, Florida, was selected as the study area for the investigation. It is located on the west coast of Florida, approximately sixty miles south of the city of Tampa. The county covers an area of approximately 570 square miles, with most of the population residing in a corridor along the coast that includes the cities of Sarasota, Venice, and Englewood (Figure 2). The local economy is fairly diversified, with trade, transportation, and utilities (20 percent); leisure and hospitality (18 percent); education and health services (16 percent); and government (12 percent) among the leaders in employment (National Association of Realtors 2010).

The area experiences a relatively high incidence of hydro-meteorological hazards, led by hurricane winds, storm surge, and lightning. The county is also exposed to riverine flooding, tornadoes, extreme heat, and wildfires (Sarasota County Emergency Management 2010).

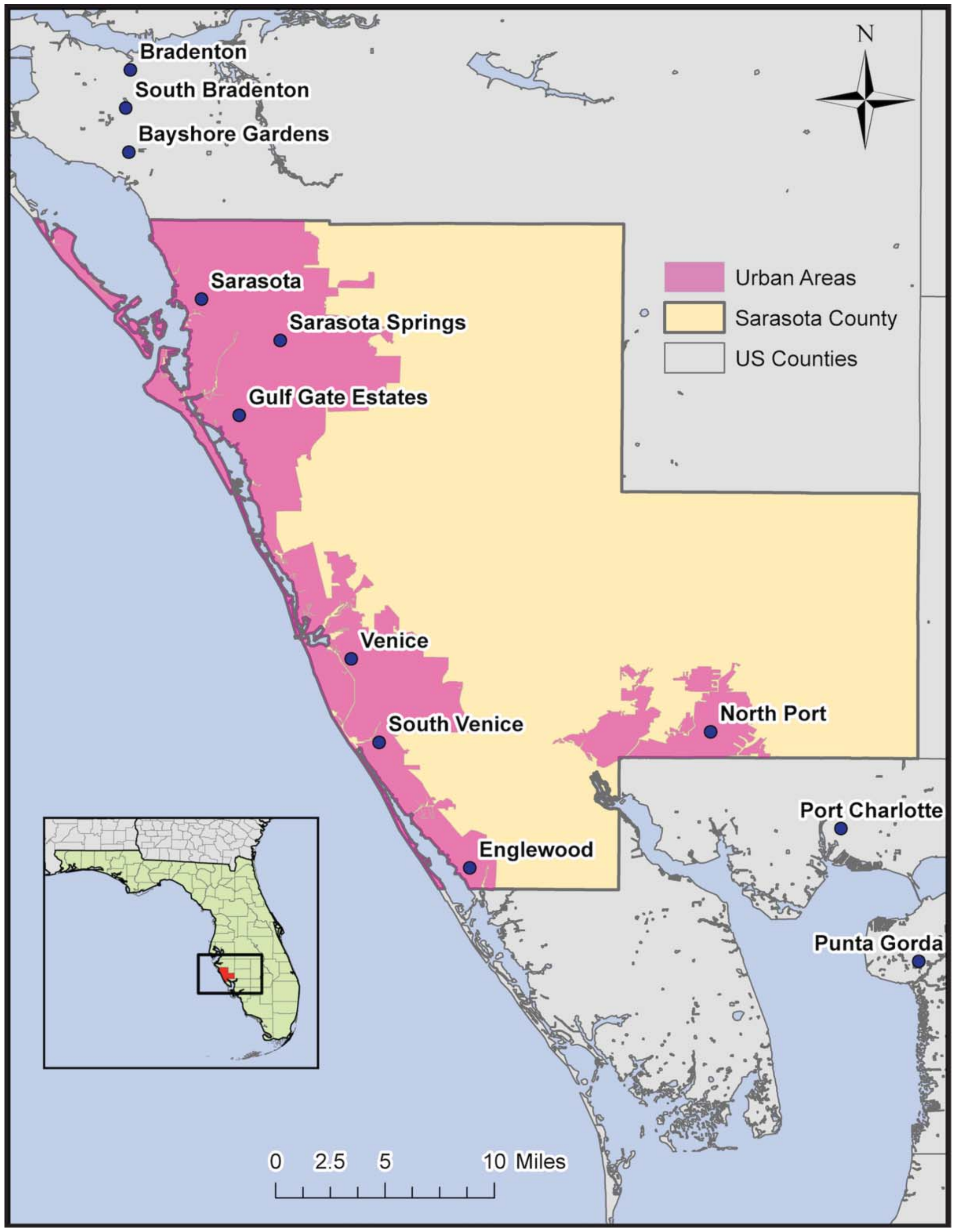

Figure 2. Sarasota County, Florida, study area. (Color figure available online.)

Selected population characteristics of Sarasota County are provided in Table 1, based on data from the 2000 U.S. Census. The county is fairly well off economically and has a relatively low minority population. Those age sixty-five and older make up a large percentage of the population, and approximately 36 percent of the residents have some kind of disability (e.g., sensory, physical, mental, employment). The high proportions of elderly and disabled suggest that the county might have a high degree of social vulnerability. This indeed was the finding in the Social Vulnerability Index for the United States (Cutter, Boruff, and Shirley 2003), computed nationwide at the county level using year 2000 census data. Sarasota ranked in the top quintile of the most socially vulnerable counties.

Methods

To assess the impact on social vulnerability ranks of changing options within each stage of index construction, Monte Carlo simulation was used as the first step in an uncertainty analysis. During each run in the simulation, a single option within each stage of index development is selected and used to compute a social vulnerability index. The vulnerability ranks from that

Table 1. Demographic profile of Sarasota County

Characteristic	Value	Percentage
Total population	325,957	—
Per capita income	$49,001	—
Below poverty	24,817	7.6
African American	13,621	4.2
Hispanic	14,142	4.3
Elderly	102,583	31.5
Female-headed households	11,565	7.7
Disabled	117,986	36.2

Table 2. Uncertainty analysis model factors

Construction stage	Options	Probability density function
Indicator set	*Base*	Discrete (0,1)
	Alternative	
Analysis scale	*Census tract*	Discrete (2,3)
	Census block group	
Measurement error	*Census 2000*	Discrete (4,5)
	Census 2000 undercounts	
Transformation	None	Discrete (6,7,8)
	Population	
	Area	
Normalization	*Linear scaling, min–max*	Discrete (9,10,11)
	Linear scaling, max value	
	Z score standardization	
Weighting	*Equal*	Discrete (12,13)
	Expert	

run are saved and the process is repeated n times. For this study, n was set to 7,168, a particular value governed by a companion sensitivity analysis study of the index construction process (Tate 2012). The overarching goal of the uncertainty analysis was to evaluate the choices commonly used in previous index studies of social vulnerability to hazards. A total of twenty-four index studies were evaluated, spanning structural designs that were deductive (Cutter, Mitchell, and Scott 2000; Montz and Evans 2001; Wu, Yarnal, and Fisher 2002; Dwyer et al. 2004; Collins, Grineski, and Aguilar 2009; Lein and Abel 2010), hierarchical (Vincent 2004; Chakraborty, Tobin, and Montz 2005; Hebb and Mortsch 2007; Flanagan et al. 2011; Mustafa et al. 2011), and inductive (G. Clark et al. 1998; Cutter, Boruff, and Shirley 2003; Rygel, O'Sullivan, and Yarnal 2006; Borden et al. 2007; Burton and Cutter 2008; Myers, Slack, and Singelmann 2008; Schmidtlein et al. 2008; Fekete 2009; Burton 2010; Finch, Emrich, and Cutter 2010; Schmidtlein et al. 2010; Tate, Cutter, and Berry 2010; Wood, Burton, and Cutter 2010). The choices of model stages and options used in the previous studies guided those selected for the uncertainty analysis, which are shown in Table 2. By and large (with the exception of measurement error), if a particular approach or parameter did not appear in a few of these studies, it was not included in the uncertainty analysis.

Two development stages shown on Figure 1 do not appear in Table 2: model structure and aggregation method. This study focused on the uncertainty associated with a hierarchical structure. The use of hierarchical models in social vulnerability analysis has been on the rise in recent years. They offer the benefit of a greater level of theoretical organization than deductive models, with less statistical complexity than inductive approaches. As such, index construction choices made in the previous hierarchical model studies more heavily influenced the design choices shown in Table 2. Regarding aggregation, previous hierarchical designs have universally employed additive aggregation. As such, this modeling stage has not been a source of epistemic uncertainty and was therefore not included in the analysis.

Each Monte Carlo simulation run begins with selection of an input sample, in which one option is chosen in each index development stage. Each stage contains either two or three options with an equal probability of selection. Discrete probability density functions (PDFs) are assigned to model the selection space. The software package SimLab version 2.2 was used to generate samples for each model run (Joint Research Centre 2011). Custom software code in VB.NET for ArcGIS 10 was developed to read the input samples from SimLab, and compute the associated vulnerability index. The set of index rankings was then saved to disk and a new simulation run initiated. For the first run in the analysis, the most frequently applied options for each model stage are combined to develop a baseline vulnerability index. Producing a reference index configuration is an important precursor to conducting robustness analysis (Cherchye et al. 2007). The baseline options for each construction stage appear in italics in Table 2.

Indicator Selection

Although the social vulnerability dimensions represented in previous indexes show similarity, the indicators selected as proxies have shown significant variation between studies. The indicators were computed using demographic variables from the U.S. Census year 2000 Summary Files 1 (SF1) and 3 (SF3). SF1 reports detailed data of population and housing

Table 3. Indicator set

Subindex	Description	Alternative
Differential access to resources	Per capita income	Persons with income below the poverty level
	Mean house value	
	Adults over age 24 with less than high school education	
	Service sector employment	Unemployed civilian laborers
	Rental housing units	
Demographic structure	Black or African American	Nonwhite and non-Anglo population
	Hispanic or Latino	People per housing unit
	Children under 5 years of age	
	Elderly age 65 and over	
	Female-headed households, no husband present	Female population
Special needs	Civilian and noninstitutionalized, over 5 years old with disabilities	
	Nursing home residents	
	Little or no English proficiency, age 5 and above	Foreign born, immigrated 1990–1999
	Housing density	
	Households with no vehicles available	

characteristics generated from responses to a short-form questionnaire, and SF3 contains statistically sampled data on population and housing units from a long-form survey sent to approximately one out of every six households. The hierarchical index is organized into three thematic subindexes: access to resources, demographic structure, and evacuation. The next task is to fill each subindex with indicators that are in alignment with its vulnerability dimension. The indicator selections are shown in Table 3. The access to resources subindex attempts to measure the amount of capital that people can draw down in a disaster. Research into livelihoods has identified five types of such capital (Wisner et al. 2004): financial, human, social, physical, and natural. The concept of socioeconomic status attempts to capture the first two of these and is the focus of this subindex. Financial resources are measured by the indicators of per capita income and mean house value, which are proxies for income and wealth. The indicators for education and service sector employment are designed to assess human capital (knowledge and skills).

If access to resources can be generalized as "what we have and what we know," the demographic structure subindex is focused differential vulnerability based on "who we are," as defined by our socialized identities. The demographic structure subindex represents dimensions of race and ethnicity, age, and gender. The indicators for this grouping embody demographic groups that have a disproportionately higher degree of hazard exposure, are more susceptible to hazard impacts, and face additional structural impediments during disaster recovery. The third subindex represents those with a variety of special needs that often lead to higher vulnerability during predisaster evacuation. The indicators selected for this category represent population groups and circumstances that demand extra attention from emergency managers. The disabled, nursing home residents, non-English speakers, and those without access to transportation are more likely to need extra assistance during predisaster evacuation. Areas with a high density of housing might require extra transportation planning so that evacuation can proceed swiftly and safely.

To examine the uncertainty of the index ranking based on the choice of indicator set, an alternative collection of indicators was applied. The subindexes of access to resources, demographic structure, and special needs were retained, but changes in the constituent indicators were made. The indicators that differ in the alternative set are also shown in Table 3.

Analysis Scale

Based on the predominance of use in the previous studies, the county level is selected to bound the analysis. Within the county, the indexes were computed at both the census tract and block group levels of aggregation to evaluate the effect of varying the scale of analysis. The goal is to test whether a given subcounty area has the same relative degree of modeled vulnerability if the analysis scale is changed. The data resolution at both the block group and tract aggregation levels is fine enough to characterize subcounty level analysis yet sufficiently coarse that socioeconomic data are available. It can be difficult to compare model results at two analysis scales, though, because it requires a common metric. Therefore, the indexes computed at the block group scale were averaged within each tract to allow for direct comparison with the tract-level indexes.

Measurement Error

In preparation for the 2000 Census, the Census Bureau planned to account for undercounts using statistical sampling techniques within a program called the Accuracy and Coverage Evaluation (ACE) Survey. However, the U.S. Supreme Court ruled in 1999 that the ACE could not be used to supplement traditional data collection approaches for purposes of apportionment. As a result, undercounts were statistically adjusted, but the revised counts were not used to adjust the 2000 census or released to the public. This latter point changed in 2002 when the detailed undercount data (ACE Revision 2) were released. Census undercount data were obtained from FairData (2003), which processed the ACE Revision 2 data set nationwide at the census block level. For use in this investigation, the data were downloaded and aggregated to the census block group and tract levels. The undercounts are based on race and ethnicity and were used to provide alternative calculations for the variables of Black, Hispanic, Asian, and Indian by adding the undercounts to the census totals. Although the undercount data set contained updated counts for sixty demographic categories, it was primarily restricted to those associated with the social vulnerability dimension of race and ethnicity. As such, there was insufficient information to enable the recalculation of any additional indicators shown in Table 3.

Data Transformation and Normalization

To include the epistemic uncertainty associated with the data transformation method, options of raw counts (none), population (percentage), and density (area) were applied. Transformation by population is achieved by dividing the indicator count by the total possible number for its category. For example, the children indicator is computed by dividing the count of children under five years old by the total population, and the renter indicator is calculated by dividing the number of rental housing units by the total number of housing units. Indicators that are not made up of counts are excluded from the population transformation. This applies primarily to economic indicators such as per capita income and median house value. The density transformation was achieved by dividing by the area of the census unit (in square kilometers). Again, non-count-based indicators are excluded from this process. To test data normalization methods, the choices of linear scaling (min–max), linear scaling (max value), and z score standardization were applied.

Weighting

For weighting the normalized indicators, two options of equal and expert weights were evaluated for the uncertainty analysis. The equal weight (w_E) for each indicator is calculated as the inverse of the total number of indicators (n):

$$w_E = \left(\frac{1}{n}\right) * 100 \tag{1}$$

For hierarchical designs, index developers must decide how to weight indicators both at the subindex and index levels. The use of equal weights at the subindex level could be problematic if all subindexes do not contain the same number of variables. Otherwise, the subindex with the greatest number of variables would receive the highest weight (OECD 2008). This study used a balanced design with respect to the number of indicators in each subindex, so this is not a concern. The second option is that of expert weights. The weights can be derived through consultation with stakeholders or those with subject matter expertise (Hoskins and Mascherini 2009) or assigned by the index developer (Vincent 2004; Mustafa et al. 2011). The expert weights used here derive from a survey of thirty-nine hazards and disaster professionals, given as part of a study examining the effect of differential weighting on social vulnerability index ranks (Emrich 2005). The experts were asked to allocate a budget of one hundred points among twenty-seven demographic indicators and to base their weights on their perceived importance in the formulaic calculation of social vulnerability. To translate the expert weights to indicators for this investigation, the following methodology was employed:

1. If an indicator has a direct match in the Emrich study, the associated weight is assigned. Unmatched indicators are assigned a weight of zero.
2. The assigned weights for all indicators are summed and the total is subtracted from 100 percent. The

Table 4. Indicator weights

Base indicator set					
Access to resources		Demographic structure		Special needs	
Indicator	Weight (%)	Indicator	Weight (%)	Indicator	Weight (%)
Income	29.2	Pop65O	22.9	NursingRes	23.0
MeanHseVal	22.5	KidsU5	21.1	Disabled	19.2
Renter	17.2	Black	20.3	NoEnglish	19.2
ServiceEmpl	15.6	Hispanic	18.3	HouseDen	19.2
Ed12Less	15.4	FemaleHH	17.3	NoVehicle	19.2
Avg. unallocated share: access to resources: 14.1%, demographic structure: 15.4%; special needs: 19.2%					
Alternative indicator set					
Access to resources		Demographic structure		Special needs	
Indicator	Weight (%)	Indicator	Weight (%)	Indicator	Weight (%)
Poverty	27.9	NonWhite	24.9	NursingRes	22.4
MeanHseVal	22.0	Pop65O	22.3	Migrant	21.8
CivilUnempl	18.7	KidsU5	20.5	Disabled	18.6
Renter	16.7	PPUnit	17.0	HouseDen	18.6
Ed12Less	14.9	Female	15.5	NoVehicle	18.6
Avg. unallocated share: access to resources: 13.6%; demographic structure: 14.8%; special needs: 18.6%					

resulting value represents the total percentage of unallocated weights.

3. The total unallocated weight total is distributed equally among all indicators and added to the assigned weights.

In this manner, weights for individual indicators were determined for the base and alternative indicator sets (Table 4). For hierarchical models employing differential weights, a decision is required to apply weights at the subindex or full-index level. The existing social vulnerability indexes provide no direction because each applied equal weighting at both levels. Applying equal weighting at the full index level is important unless there is an important justification otherwise (Villa and McLeod 2002), the approach taken in hierarchical index development elsewhere (Hoskins and Mascherini 2009). As such, this precedent is followed; differential weights (from the Emrich study) were applied at the subindex level and equal weights at the full-index level.

Results

Figure 3 presents the baseline index for the hierarchical model. Census tracts labeled as high vulnerability represent the top quintile in the baseline ranking. The bottom quintile signifies low vulnerability, with the remaining 60 percent of tracts assigned a moderate vulnerability ranking. This classification is intended to highlight areas of high and low vulnerability that are likely to be of interest in a typical vulnerability assessment. Four high social vulnerability clusters are evident in and around the cities of Sarasota, Venice, Englewood, and North Point. Groupings of low vulnerability census tracts are found along the northern coast and in areas east and south of Sarasota Springs. With the index computed and mapped, previous studies have generally proceeded to describe the geographic distribution of social vulnerability, identify hot spots, determine which input variables contribute the most to vulnerability, perform statistical analysis with ancillary data, and perhaps suggest strategies for use of the index in vulnerability reduction.

The vulnerability analysis in previous studies typically ends with the presentation and discussion of the baseline index. Left unaddressed, however, is the reliability of the index ranks. Could minor changes in the index construction approach yield a markedly different set of rankings? Without performing an uncertainty analysis, this question cannot be adequately answered. A robust index is one that is both analytically and statistically sound (Birkmann 2006). In the context of uncertainty, this means that minor changes in construction

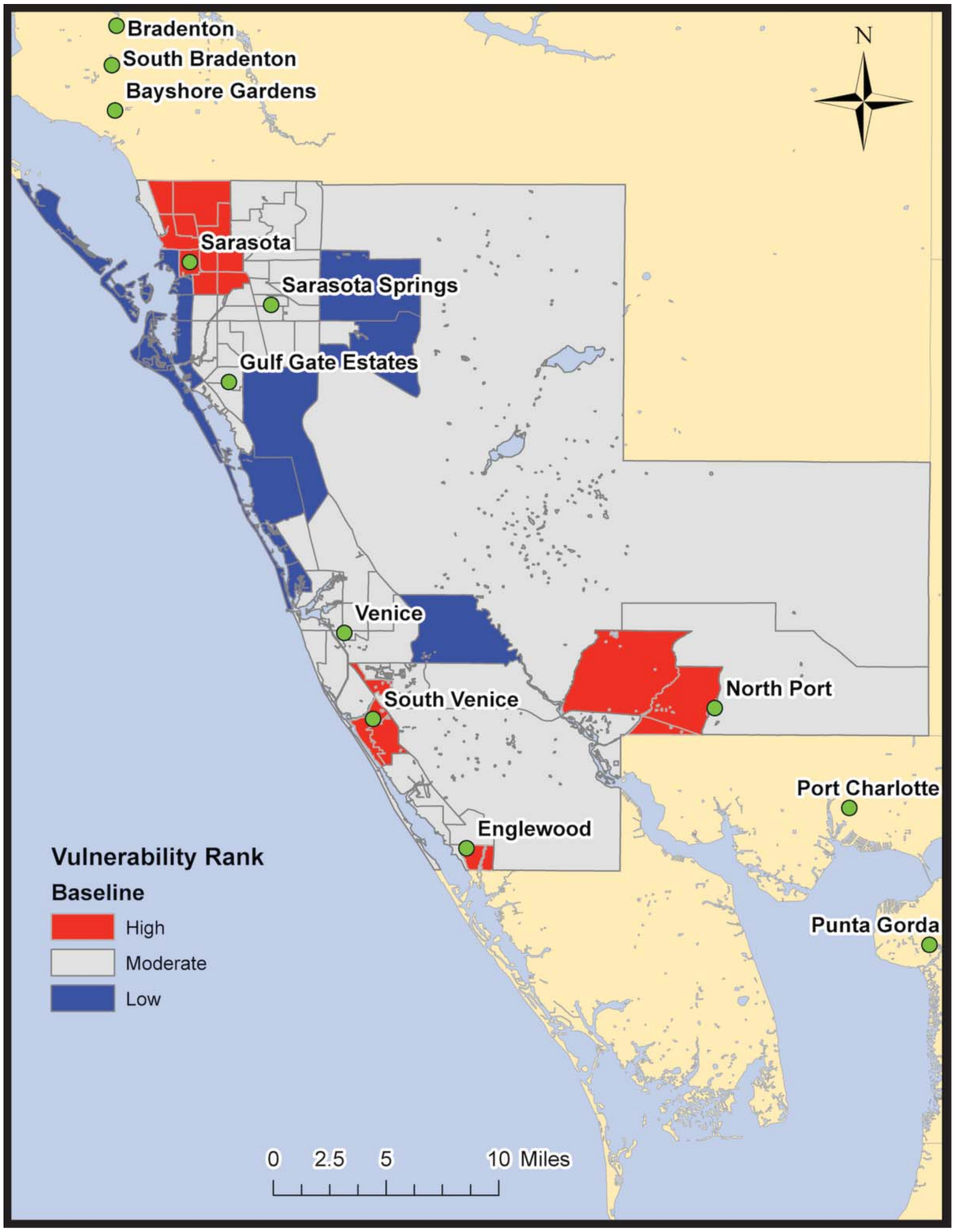

Figure 3. Baseline index. (Color figure available online.)

methodology do not produce large variations in output ranks. A lack of robustness opens the door for decisions made based on index rankings to be called into question. Instead of the typical output of a single rank for each census tract, adding Monte Carlo simulation to the index development process generates a frequency distribution of social vulnerability ranks for each tract. Statistics describing the distributions can then be used to help evaluate the robustness of the index. Three statistics of particular use are confidence intervals (CIs) to assess uncertainty magnitude, the median to evaluate bias, and the coefficient of variation to assess index precision (Figure 4). The application of these statistics in the uncertainty analysis is explored in the following sections.

Uncertainty Magnitude

The range of the frequency distribution measures the magnitude of variation in index ranks, and is computed to address this research question: How much uncertainty is associated with the social vulnerability index? Five percent of the observations are removed from the tails of the distribution at each census tract to produce the 90 percent CI (Figure 4). In this manner, the presence of outliers does not inordinately distort

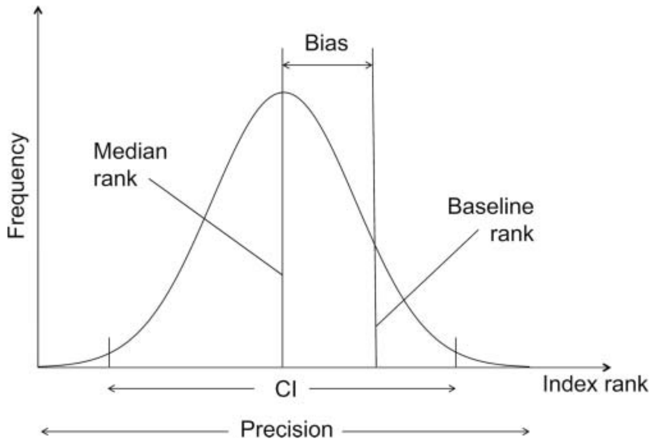

Figure 4. Uncertainty analysis statistics. CI = confidence interval.

the uncertainty findings. Previous studies of index uncertainty have displayed the rank and CIs using box plots or tables (Hoskins and Mascherini 2009; Saisana and Saltelli 2010). This approach works well when the enumeration units are recognizable, such as country or university names, but when the enumeration units are large in number or are identified cryptically as is the case for census tract numbering, this presentation style can become cluttered and confusing. Furthermore, the ability to compare the spatial distributions of rank and CI is lost. These features suggest the use of a mapping solution for the display of index uncertainty.

The cartographic representation of uncertainty has received increasing attention in the literature in recent years. Techniques for visualizing attribute uncertainty include color saturation, crispness, transparency, texture, interactivity, animation, and dimensionality (MacEachren et al. 2005; Bostrom, Anselin, and Farris 2008). Here, the use of crispness is employed as it is both an intuitive and widely applied uncertainty visualization technique (Pang 2008). The result is provided as Figure 5. Social vulnerability based on the median rank is categorized into three classes based on quintiles and symbolized using the same color scheme as in Figure 3. Epistemic uncertainty, as represented by the 90 percent CI, is displayed using two classes of high (top quintile) and moderate (remaining 80 percent). High uncertainty values are symbolized with a lack of crispness.

Two classes for the 90 percent CI were selected instead of three to reduce the complexity of the bivariate mapping. Tracts just west of North Port have a high degree of vulnerability but also a wide CI. Without the uncertainty analysis, these areas might be prioritized for hazard mitigation, but the finding of high uncertainty suggests that such a decision warrants further consideration. Meanwhile, the stretch of low vulnerability tracts along the northern coast is validated. With the bivariate mapping approach, map users are presented with additional information that might inform end use of the vulnerability index. For areas with low uncertainty, decision makers can proceed to implement vulnerability reduction strategies with a greater degree of confidence that the index rankings are reliable. High uncertainty areas signify locations where additional information might be required.

Lost in the ordinal classification of the CI are the underlying values. The quintile with the greatest uncertainty has a range of fifty-one to seventy rank positions. This is an astounding degree of variation given that there are only eighty-three census tracts in total. Conversely, the quintile with the lowest uncertainty ranges from two to twenty-four positions. These are the best performing tracts in terms of model robustness, yet some still exhibit very large degrees of variability. Across all tracts, the mean value of the 90 percent CI is thirty-five rank positions. Census tracts in the high CI uncertainty quintile appear to be geographically spread across the county. A few coincide with locations of high vulnerability, but most are found in areas of moderate social vulnerability. The Pearson correlation coefficient between vulnerability rank and 90 percent CI is 0.16, suggesting a weak positive linear association between high rank (low vulnerability) and the 90 percent CI. The global Moran's *I* test was performed to assess this association based on spatial autocorrelation. Spatial autocorrelation tests are useful for analyzing the strength and direction of spatial dependency. The Moran's *I* bivariate coefficient of 0.04 suggests a spatial relationship so weak that it can be described as random.

Uncertainty Bias

The baseline rankings are the most common output of the index development process, but the subjective choices made by the index developer during the construction process leave open the possibility that the baseline is a statistically biased estimation of vulnerability. This begs these questions: How does the baseline rank perform compared to alternative index configurations? Is it an outlier or a more central measure? To address these questions, the median rank based on the Monte Carlo distribution is evaluated:

> As there is no universally accepted model to construct a composite indicator, different scenarios (simulations) need to be considered. Hopefully, the picture provided by

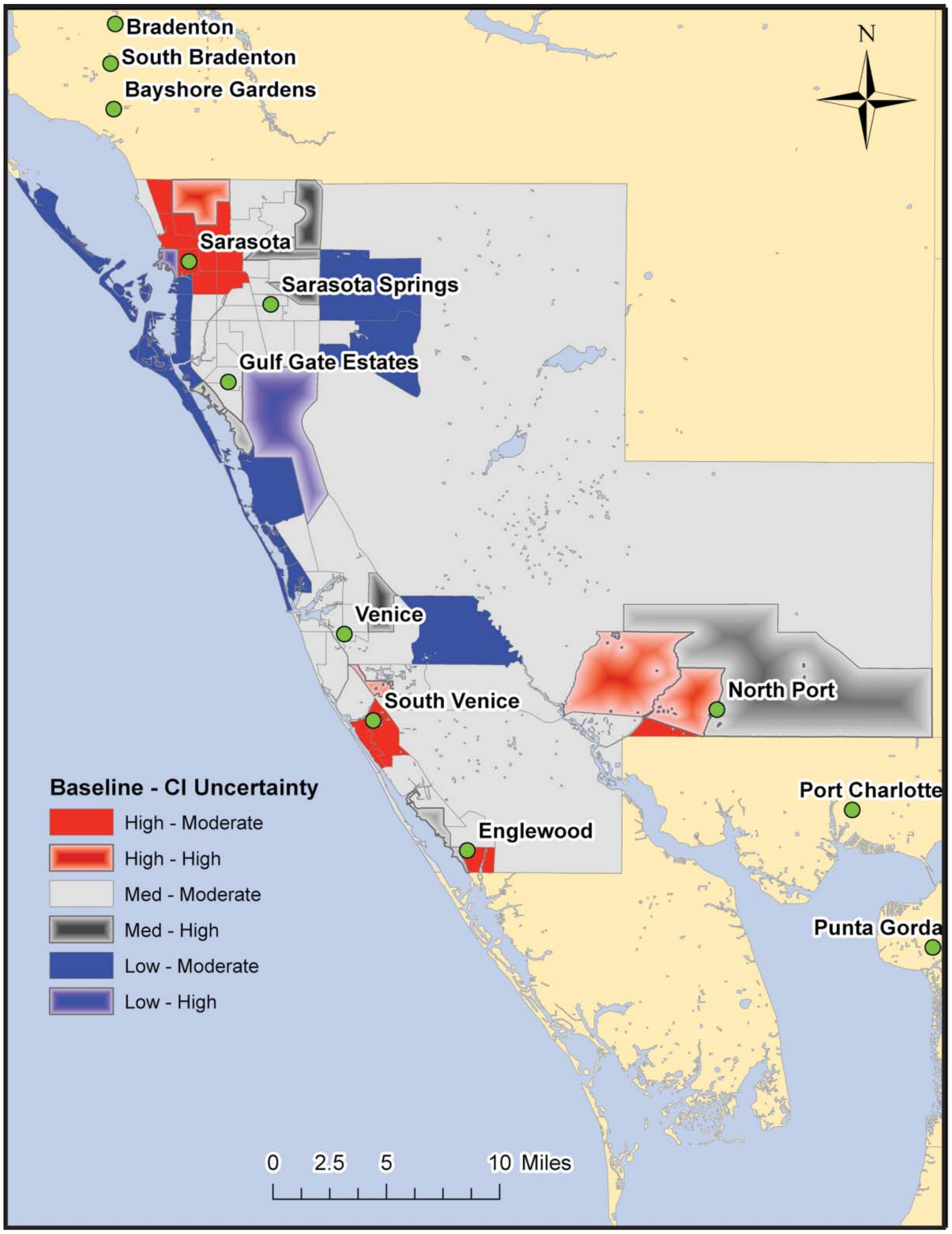

Figure 5. Bivariate vulnerability and 90 percent confidence interval (CI). (Color figure available online.)

> a composite indicator will resemble the "median" of the simulated scenarios, in order to be considered representative of the greater space of inferences. (Saisana and Saltelli 2006, 8)

A set of baseline ranks that are close to the median values is a sign of a robust index, whereas baseline values hovering near the confidence limits indicate a severely biased estimation. The statistical bias of the baseline index is computed as the deviation between the baseline and median rankings. The results are mapped in Figure 6. The census tracts symbolized in purple represent places where the median rank exceeds the baseline by at least five positions. These are *potential* locations of Type I error (false positive): tracts identified as vulnerable by the baseline model that might not be. In contrast, the tracts symbolized in brown are locations where the baseline rank exceeds the median rank by at least five positions. These are potential locations of Type II error (false negative): tracts not identified as vulnerable by the baseline index that are. The choice of five rank positions as an outlier threshold is fairly arbitrary but serves to illustrate the point. All told, one third of the tracts are outliers. If the threshold is increased to ten rank positions, then 15 percent of the tracts qualify as outliers. It is left to the index developer

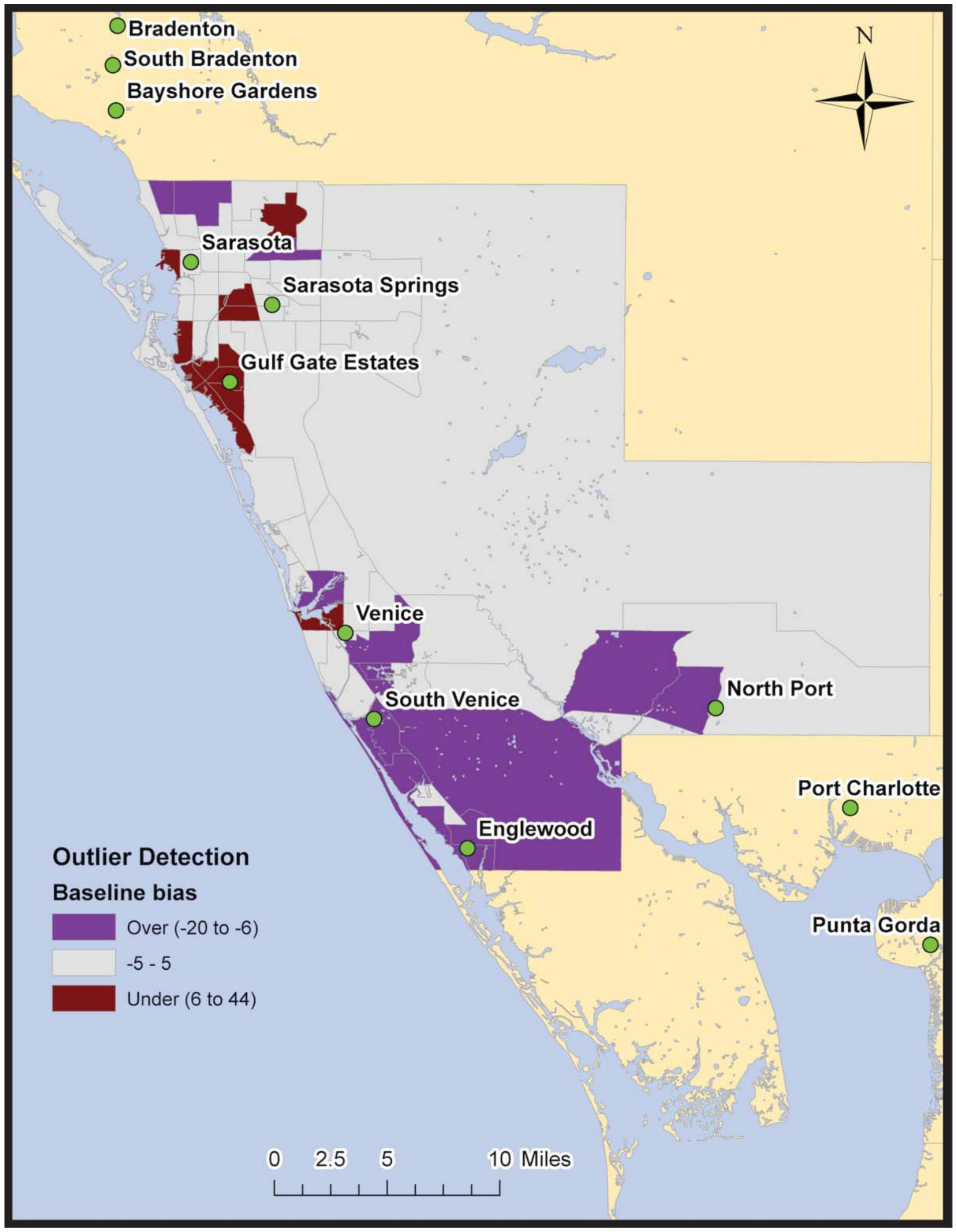

Figure 6. Baseline bias. (Color figure available online.)

and users to determine what threshold should be applied, but computation and use of the median provides useful information with which to judge the bias of the index. To eliminate bias and blunt criticism of subjectivity, one option for index developers is to replace the baseline ranks with the median ranks and present the result as the index.

Uncertainty Precision

Just as the median provides an alternative measure of the vulnerability rank, the coefficient of variation offers another way to assess uncertainty. The 90 percent CI as a measure of uncertainty is a coarse metric. It captures the range of potential ranks yet is incapable of measuring the variability of ranks within it. A tract with a few simulated rankings near the confidence limits but with most near the median (low uncertainty) would be assigned the same degree of uncertainty as one with ranks evenly spread throughout the CI (high uncertainty). Computed as the standard deviation divided by the mean, the coefficient of variation (CV) helps to differentiate these cases. As a rule of thumb, CV values below 12 percent signify a high degree of precision in the vulnerability rank distribution, and values above 40 percent are suggestive of low precision (ESRI 2011).

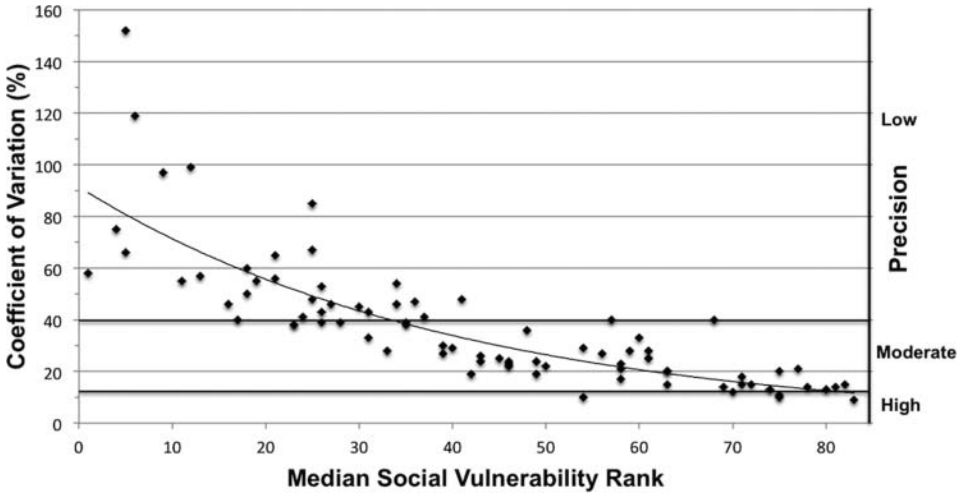

Figure 7. Index precision vs. median rank.

These thresholds are illustrated in Figure 7, which plots the CV against the median rank.

The relationship between median vulnerability and CV precision is much stronger than that found earlier between baseline vulnerability and CI magnitude. The Pearson correlation coefficient between the median and CV is –0.71, suggesting a negative association between vulnerability rank and index variability. In other words, the higher the vulnerability, the greater the index uncertainty. This relationship is apparent in Figure 7 and echoes findings from previous studies (Saisana 2008; Aguña and Kovacevic 2011) of a positive association between poor index performance and high variability. Social vulnerability indexes are typically used to identify areas of high vulnerability, but it is precisely in these places where the precision of the model is at its nadir. This suggests a potential alternative use of the index not as an approach to identify high-vulnerability areas but as a screening tool to eliminate low-vulnerability areas from consideration. The vulnerability of the remaining areas could then be further investigated using subsequent uncertainty analysis, sensitivity analysis, or nonquantitative methods such as interviews and participatory approaches.

Conclusions

Hazard mitigation initiatives have traditionally focused on physical and economic aspects of vulnerability. In recent years these efforts have begun to more seriously consider social dimensions. Accordingly, there has been increased interest in indexes as a metric measure and mapping social vulnerability to hazards, and social vulnerability indexes have begun to appear in U.S. state and federal mitigation planning documents. The academic literature has served as a foundation for this trend, seeking to improve vulnerability measurement through the exploration of different model structures, indicator sets, and weighting approaches. Yet as a research community, we know far too little about the impact of these modeling decisions on the resulting social vulnerability metrics.

The goal of this research has been to apply the common choices made in previous vulnerability studies to quantify and visualize uncertainty for a hierarchical social vulnerability index. Six sources of epistemic uncertainty were considered in a Monte Carlo simulation: indicator selection, scale of analysis, measurement error, data transformation, normalization, and weighting. Not all stages of index construction are necessarily sources of uncertainty for all indexes, though. For example, an index focused on evacuation vulnerability might have less epistemic uncertainty associated with the indicator selection process than a generalized vulnerability index. Likewise, the scale of analysis might not be an uncertain factor for an index explicitly designed for use at a particular administrative scale. In cases such as these, index uncertainty would be lower than predicted in this all-encompassing analysis.

Overall, the analysis characterized the index as imbued with a high magnitude of uncertainty, statistically biased, and less precise in areas of high vulnerability. What index construction stages are driving the uncertainty and why are the rankings in some vulnerable areas robust and others not? Sensitivity analysis is required to fully address these questions and is the focus of a parallel research effort (Tate 2012). The findings suggest that it is the weighting stage that is the key driver of uncertainty for the hierarchical model in the

Sarasota study area. With this additional knowledge, methodological scrutiny can be devoted to choosing the weighting method that best measures vulnerability based on the purpose of the index.

The findings presented herein form part of a larger study that considers additional index types (deductive and inductive) and study areas (Norfolk, Virginia, and Nueces County, Texas). The work produced a total of nine uncertainty analyses, which largely confirmed the findings presented herein for the hierarchical index in Sarasota County, Florida (Tate 2011). Collectively, the results of the uncertainty analysis should give pause to those who seek to uniformly apply social vulnerability index rankings to hazard mitigation decision making. The modeling decisions made during index construction really do matter. Too often, previous vulnerability index studies have made subjective modeling decisions with little or no stated justification. Subjectivity is not inherently a bad thing, and some degree of its use is required in any modeling effort. But the effects of subjective choices on the output index should be assessed prior to employing it in hazard mitigation activities such as project prioritization, resource allocation, and advocacy. The uncertainty assessment and visualization methods presented in this article can be used toward that end. Furthermore, the detailed examination of modeling decisions required for the uncertainty analysis adds transparency to the index construction process.

The quantification of social vulnerability is an important and long overdue addition to the hazard mitigation planning process. The heart-wrenching imagery of the aftermath of Hurricane Katrina attests to the peril of ignoring social dimensions of hazards vulnerability. The use of highly uncertain metrics might not bear the vulnerability reduction fruit that is hoped for, however. The addition of uncertainty analysis to the index construction process is therefore an important step toward improving the quality of the next generation of social vulnerability indexes. Epistemic uncertainty is associated with all vulnerability models. The lack of its assessment and portrayal does not deny its existence.

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